

CIS 6270

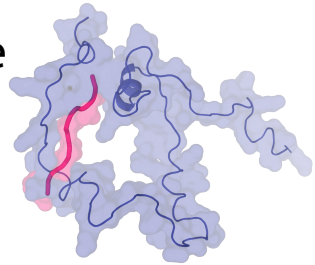
Module 1: Introduction and Math Foundations

Lecture 1.1: Probability Foundations

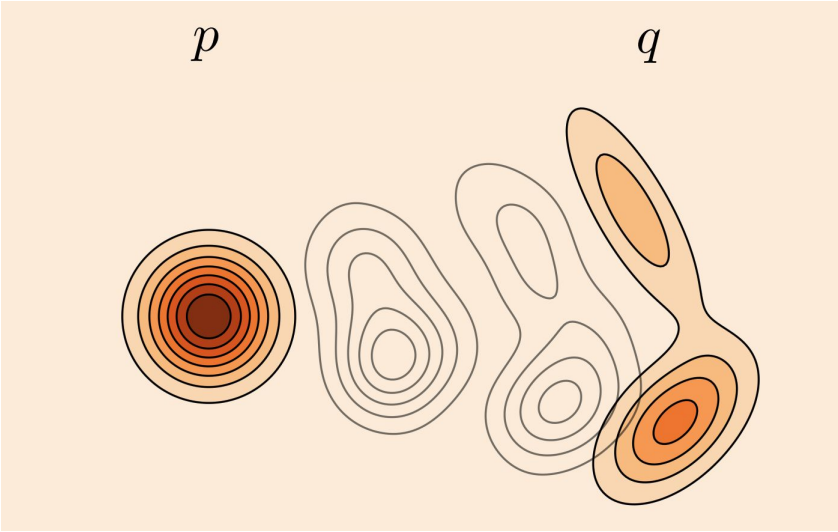
Tue, Aug 25, 2026

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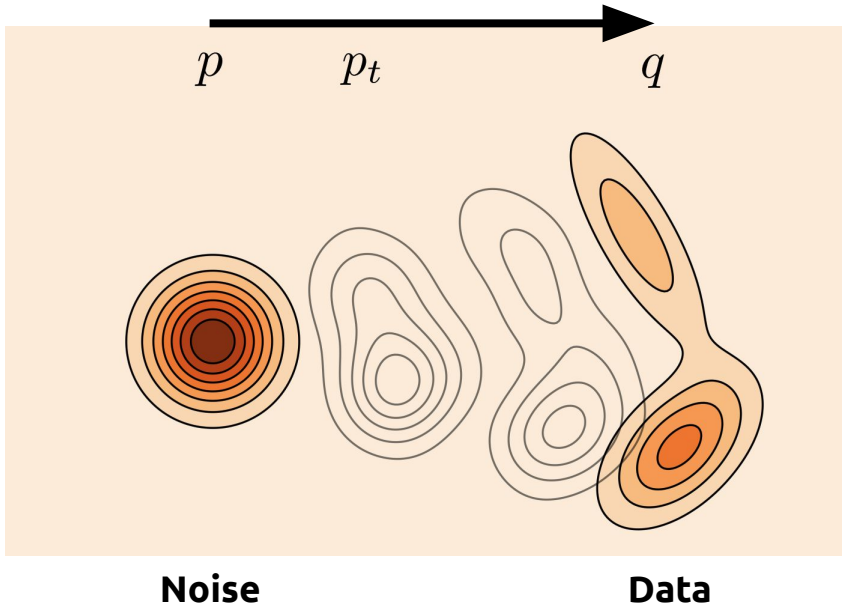


What is a generative model?



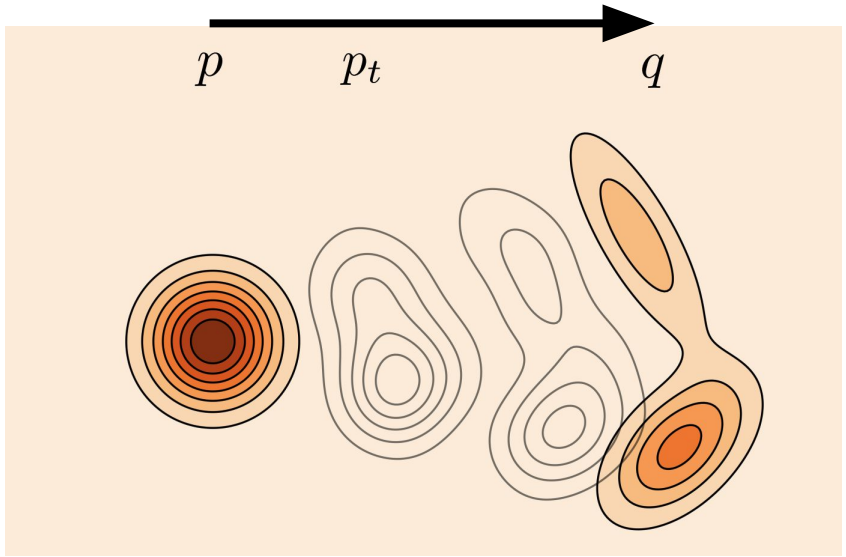
What is a generative model?

- A learned process that transforms a SIMPLE distribution to the DATA distribution.



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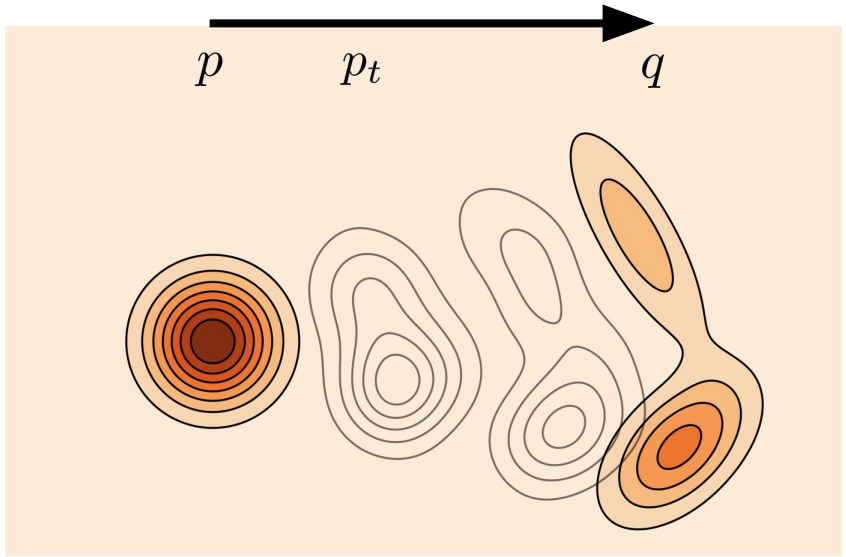
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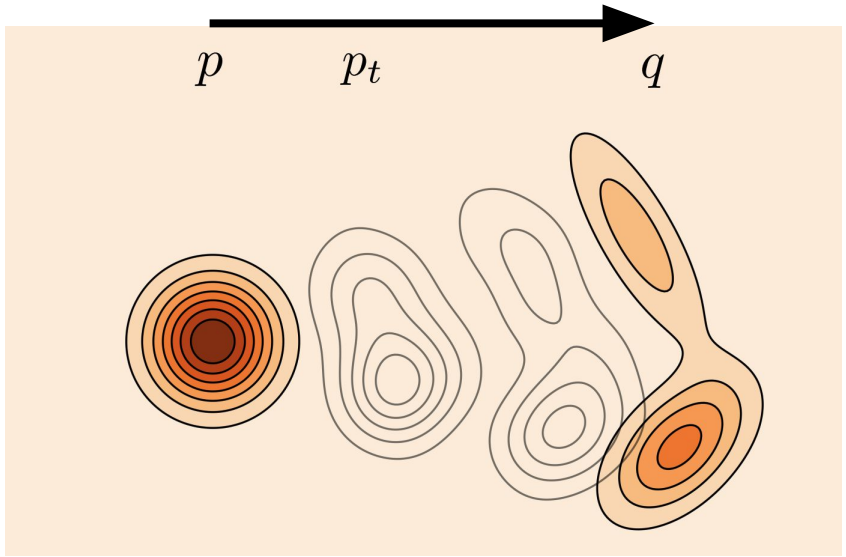
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Intermediate distribution

$$p_0(\mathbf{x}) = p(\mathbf{x}), \qquad p_T(\mathbf{x}) = q(\mathbf{x})$$

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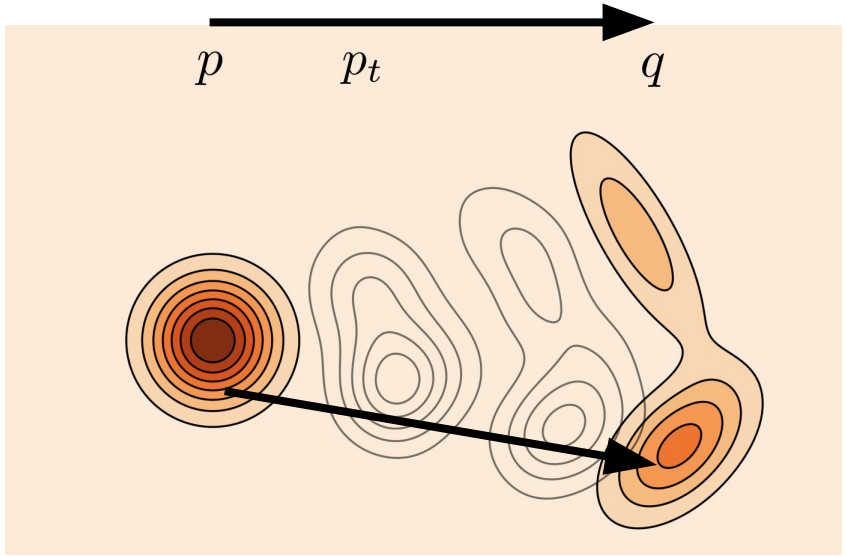


$$p_0(\mathbf{x}) \xrightarrow{p_t(\mathbf{x})} q(\mathbf{x})$$

$$\mathbf{x}_T \sim q(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$$

For generation:

- Sample x_0 ~ p, then evolve it along the learned path p_t until x_T ~ q.



$$p_0(\mathbf{x}) \xrightarrow{p_t(\mathbf{x})} q(\mathbf{x})$$

$$\mathbf{x}_T \sim q(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$$

Two types: Continuous vs. Discrete.

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$$\textbf{Continuous:} \quad x_i \in \mathbb{R}$$

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Continuous

$$\begin{bmatrix} 0.31 \\ -1.27 \\ 2.85 \end{bmatrix}$$

Discrete

$$\begin{bmatrix} \texttt{CAT} \\ \texttt{SAT} \\ \texttt{MAT} \end{bmatrix}$$

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$$\textbf{Continuous:} \quad x_i \in \mathbb{R}$$

$$\textbf{Discrete:} \quad x_i \in \mathcal{V}$$

Continuous

$$\begin{bmatrix} 125 \\ 180 \\ 200 \end{bmatrix}$$

RGB Pixel

Discrete

$$\begin{bmatrix} \texttt{A} \\ \texttt{C} \\ \texttt{G} \\ \texttt{T} \end{bmatrix}$$

DNA

So instead of moving continuously through Euclidean space, we're moving through combinations of symbols

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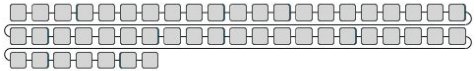
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DNA

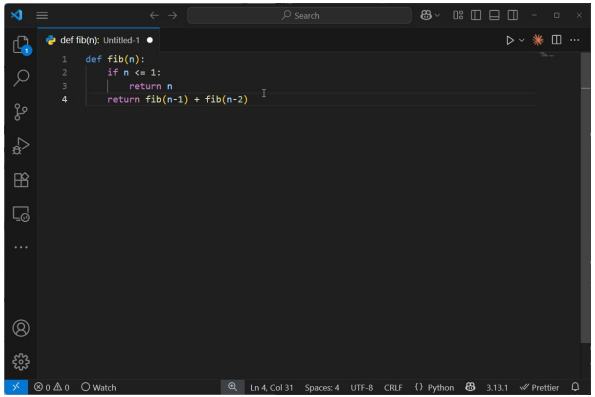
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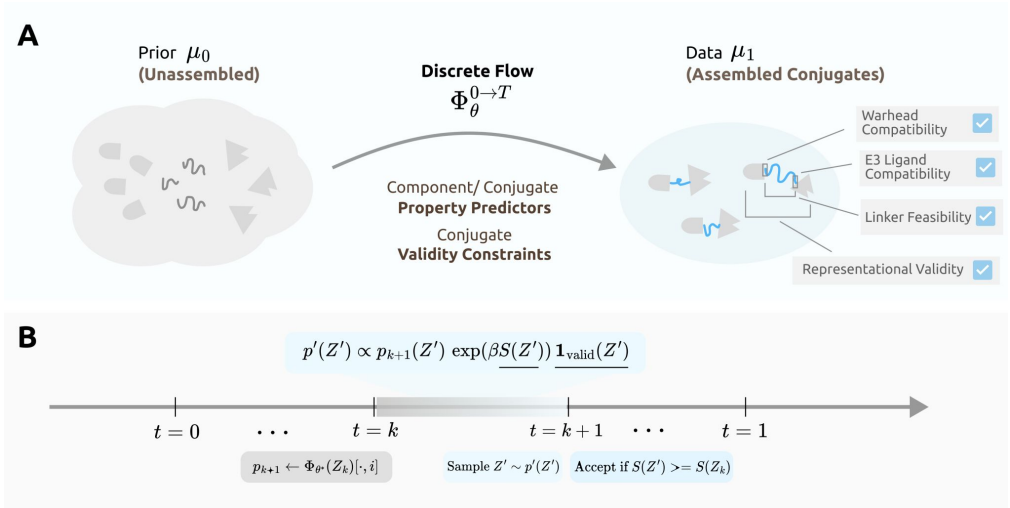


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Why **Discrete** Generative Models?

- Many real-world data are sequences of symbols
- Outputs must satisfy discrete constraints



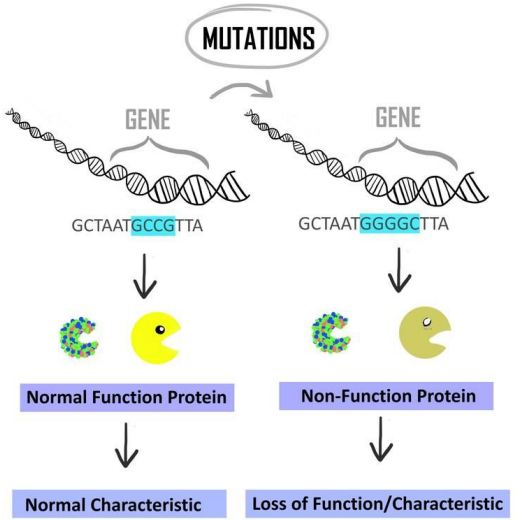
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- Small edits can produce large semantic changes

A – Z Similar words with Different Meanings

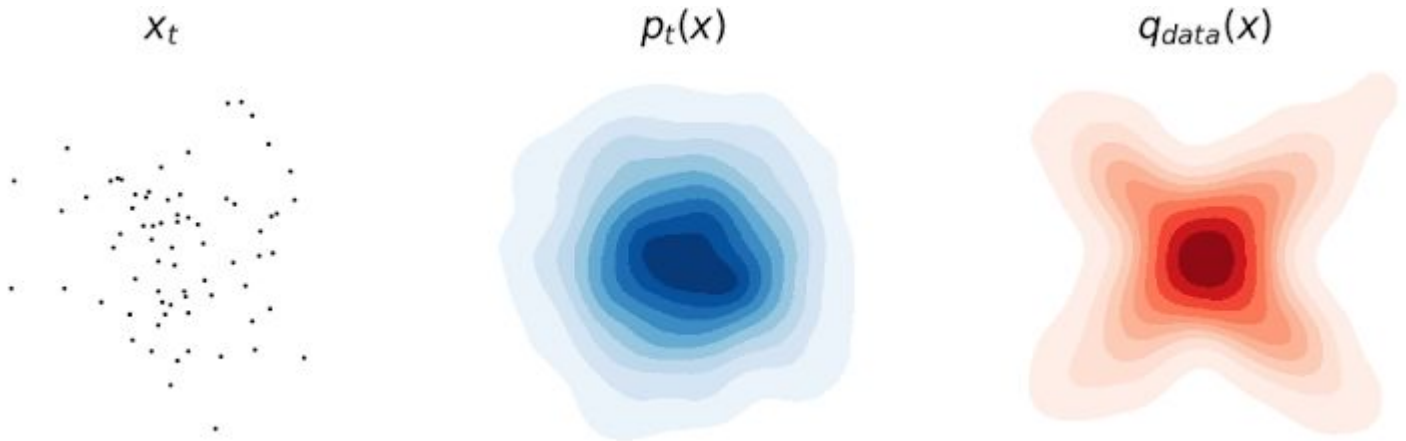
Word	Meaning	Word	Meaning
Abject	something bad	Industrial	Commercial
object	Material	Industrious	Hard-working
Casual	Relaxed	Lose	Misplace
Causal	Permanent	Loose	Baggy
Foul	Wicked	Marry	Wed
Fowl	Bird	Merry	Cheerful
Heal	Fill up	Vain	Useless
Heel	The lower part of the feet below the ankle	Vein	Blood vessels

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Why **Discrete** Generative Models?

- Many real-world data are sequences of symbols
- Outputs must satisfy discrete constraints
- Small edits can produce large semantic changes
- New stochastic processes are needed for generation



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- The state changes by discrete jumps, but those jumps occur in continuous time.

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$$\boxed{x_t \in \mathcal{V}^L, \qquad t \in [0, T]}$$

$$\mathcal{V} = \{A, C, G, T\}, \qquad \mathcal{V} = \{\texttt{cat}, \texttt{dog}, \texttt{bird}\}, \qquad \mathcal{V} = \{\texttt{<MASK>}, A, \ldots, Z\}$$

$$0 \longrightarrow 0.13 \longrightarrow 0.81 \longrightarrow 1.42 \longrightarrow T$$

Specifically, we will operate primarily in **discrete space** but **continuous time**

- Beginning with the initial distribution $p_0(x)$, as continuous time t evolves from 0 to T, the probability distribution changes until the final distribution $p_t(x)$ approximately equals the target distribution q(x).

$$x_t \in \mathcal{V}^L, \qquad t \in [0, T]$$

$$p_0(\mathbf{x}) \xrightarrow{t\in[0,T]} p_T(\mathbf{x}) \approx q(\mathbf{x}), \qquad \mathbf{x} \in \mathcal{V}^L$$

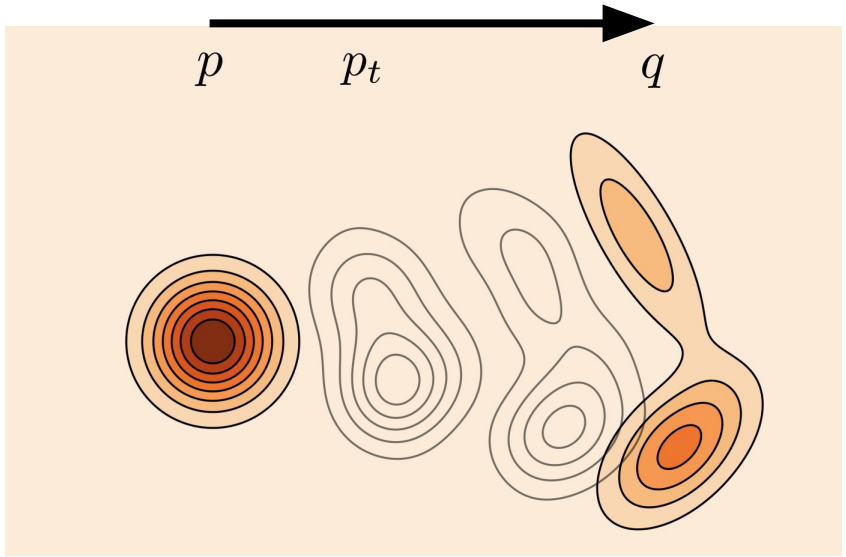
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- Discrete state spaces and probability distributions
- Markov chains, CTMCs, and master equations
- Autoregressive sequence models
- Discrete diffusion models
- Discrete flow matching
- Schrödinger bridges and controlled stochastic processes

Remember what a generative model is.

- A learned process that transforms a SIMPLE distribution to the DATA distribution.

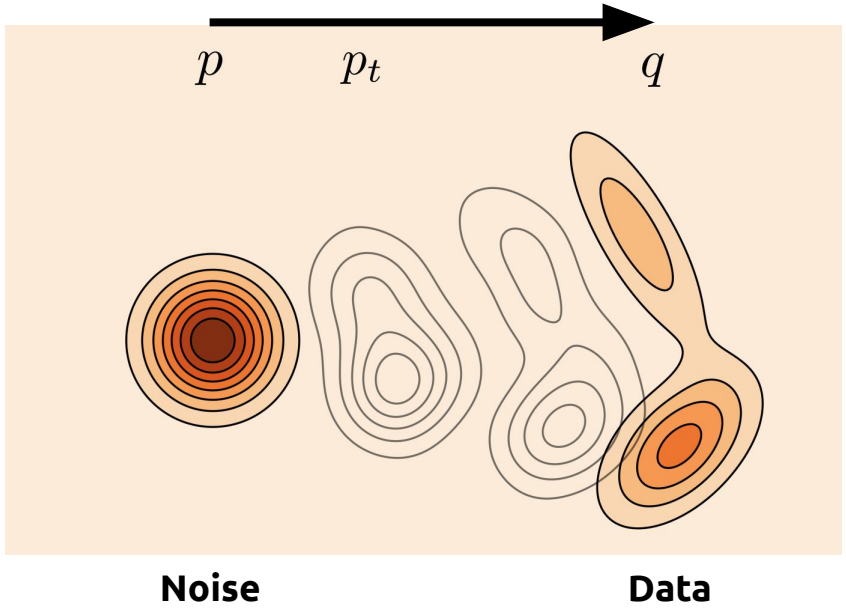


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A probability distribution cannot assign probabilities until we specify every possible value the variable could take



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State Space

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$\mathscr{X}$ is the state space containing every length-L sequence constructed from tokens in the vocabulary $\mathscr{V}$.

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For a protein sequence with 20 possible amino acids at every position:

$$|\mathcal{X}| = 20^L$$

The random variable X follows the distribution p_X, and x is one realized value of X.

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We use lowercase x for both the random object and its realized value

Discrete Random States

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$$0.4 + 0.1 + 0.2 + 0.3 = 1$$

Therefore, this is a valid probability distribution.

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The probability of an event is the sum of the probabilities of all states inside it:

$$\mathbb{P}(X \in A) = \sum_{x \in A} p_X(x)$$

The probability that X belongs to the event A equals the sum of p_X over every state x contained in A.

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$$\mathbb{P}(X \in A) = p_X(\mathrm{AA}) + p_X(\mathrm{AC}) = 0.4 + 0.1 = 0.5$$

So the first token is A with probability 0.5.

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The area under the density over a region is a probability. A density may even exceed 1 when it is concentrated over a sufficiently narrow region!

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Expectation

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For biological generation, we may want to know:

- The average number of a particular nucleotide
- The expected length of a sequence
- The expected binding score
- The average charge of generated peptides
- The expected training loss

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The function is written as:

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$$\mathbb{E}_{X \sim p_X}[h(X)] = \sum_{x \in \mathcal{X}} p_X(x) h(x)$$

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Expectation is a probability-weighted average. States with higher probability contribute more strongly.

Continue with the previous distribution:

$$
\begin{aligned}
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Define h(x) as the number of A tokens in sequence x:

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For 1000 generated sequences, we would expect approximately 1100 total adenines across the two positions.

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The mean is often denoted by:

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The discrete and continuous definitions perform the same weighted averaging operation. A sum is used for discrete states, while an integral is used for continuous states.

Expectation distributes across sums and constant coefficients

$$\mathbb{E}[a\,f(X) + b\,g(X)] = a\,\mathbb{E}[f(X)] + b\,\mathbb{E}[g(X)]$$

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Linearity requires no independence assumption. It applies even when the quantities being added are statistically dependent.

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Consider two random variables:

$$X_1 = \begin{cases} -1 & \text{with probability } \frac{1}{2}, \\ 1 & \text{with probability } \frac{1}{2}, \end{cases} \qquad X_2 = \begin{cases} -10 & \text{with probability } \frac{1}{2}, \\ 10 & \text{with probability } \frac{1}{2}. \end{cases}$$

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Positive and negative deviations cancel when averaged

$$\mathbb{E}[X - \mu_X] = \mathbb{E}[X] - \mu_X = 0$$

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$$\begin{aligned} \mathbb{E}[X^2] &= 0^2(0.2) + 1^2(0.5) + 2^2(0.3) \\ &= 0 + 0.5 + 1.2 \\ &= 1.7. \end{aligned}$$

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$$\begin{aligned} \mathbb{E}[X^2] &= 0^2(0.2) + 1^2(0.5) + 2^2(0.3) \\ &= 0 + 0.5 + 1.2 \\ &= 1.7. \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= 1.7 - (1.1)^2 \\ &= 1.7 - 1.21 \\ &= 0.49. \end{aligned}$$

Remember, standard deviation returns to the original units

$$\sigma_X = \sqrt{\mathrm{Var}(X)}$$

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While variance describes fluctuations of one variable, covariance describes how two variables fluctuate together.

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- Variance is covariance with itself: $\operatorname{Var}(X) = \operatorname{Cov}(X, X)$

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Let X represent hydrophobicity and Y represent a membrane-binding score.

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The covariance is positive because hydrophobicity and binding score increase together.

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The entry in row i and column j is

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$$

For two variables,

$$\Sigma = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) \end{bmatrix}$$

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$$\Sigma = \begin{bmatrix} 4 & 3 \\ 3 & 9 \end{bmatrix} \quad \Longrightarrow \quad \Sigma^\top = \begin{bmatrix} 4 & 3 \\ 3 & 9 \end{bmatrix} = \Sigma$$

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The covariance matrix equals its transpose because swapping the order of the variables does not change their covariance

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$$X_0 \sim \mathcal{N}(0, I)$$

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The signal covariance shrinks as α_t decreases, while isotropic (same variance in every directions) noise covariance grows with σ_t^2

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We therefore need a distribution over several variables simultaneously.

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Every joint probability p(x,y) is nonnegative, and the sum over all possible values of x and y equals one.

Suppose X indicates whether a motif is present and Y indicates whether binding is high

	$Y = \text{high}$	$Y = \text{low}$	$p(x)$
$X = \text{motif}$	0.36	0.24	0.60
$X = \text{no motif}$	0.08	0.32	0.40
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The four joint probabilities sum to one:

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But what if we only care about whether the motif is present

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Marginal Distriution

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It describes one variable on its own after averaging over all possible values of the other variables

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We sum over every possible binding outcome:

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But now suppose we know that the motif is present.

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$$\begin{aligned} p(\text{high} \mid \text{motif}) &= \frac{p(\text{motif}, \text{high})}{p(\text{motif})} \\ &= \frac{0.36}{0.60} \\ &= 0.60. \end{aligned}$$

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Recall, the overall probability of high binding is 0.44.

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Conditioning restricts attention to the outcomes consistent with the observed information and then renormalizes their probabilities.

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And this leads us to Bayes' rule!

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What is the probability that a sequence contains the motif after observing high binding?

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$$\boxed{p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)}}$$

The posterior probability p(x given y) equals the likelihood p(y given x) multiplied by the prior p(x), divided by the evidence p(y).

What is the probability that a sequence contains the motif after observing high binding?

$$\begin{aligned} p(\text{motif} \mid \text{high}) &= \frac{p(\text{high} \mid \text{motif})p(\text{motif})}{p(\text{high})} \\ &= \frac{(0.60)(0.60)}{0.44} \\ &= \frac{0.36}{0.44} \end{aligned}$$

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Before observing binding, the motif probability is 0.60.

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This is the law of total probability.

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In the numerical example,

$$p(\text{high} \mid \text{motif}) = 0.60$$

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The variables are therefore dependent.

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Repeated application of the product rule gives the probability of a complete sequence

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The joint probability of the sequence x_1 through x_L equals the product over positions i of the probability of token x_i given all preceding tokens.

This factorization is exact for every joint distribution.

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Autoregressive models, like GPT, parameterize these conditional probabilities and generate one token at a time.

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The probability of the clean sample x_0 given the noisy sample x_t equals the forward corruption probability $p(x_t$ given $x_0)$ times the clean-data prior $p(x_0)$, divided by the noisy marginal probability $p(x_t)$.

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During training, we sample clean data x_0, corrupt it into x_t using the known forward distribution $p(x_t \mid x_0)$, and train the model to predict x_0 or the added noise from x_t. Across many examples, the model learns an approximation to the reverse distribution $p(x_0|x_t)$.

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The noisy distribution $p(x_t)$ combines every possible clean sample x_0 with the noise process $p(x_t|x_0)$. During training, we approximate this automatically by sampling x_0 from the dataset and corrupting it to obtain x_t.

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A denoising model approximates information about this reverse conditional distribution.